

## Scalability & Future Plan

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|----------------------|------------------------------------------|
| <b>Date</b>          | 04 July 2026                             |
| <b>Team ID</b>       | XXXXXXX                                  |
| <b>Project Name</b>  | Smart-Lender-Loan-Eligibility-Prediction |
| <b>Maximum Marks</b> | 1 Mark                                   |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                                                   | Impact                                                  | Priority to Address<br>(High / Medium / Low) |
|------|----------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------|
| 1    | The application runs on a local Flask server and is not deployed on the cloud.               | Cannot be accessed by multiple users over the internet. | High                                         |
| 2    | The application uses a CSV dataset and does not store user prediction history in a database. | No persistent storage or record management.             | Medium                                       |
| 3    | The system does not include user authentication or authorization.                            | Limited security for accessing the application.         | High                                         |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                               | Proposed Upgrade / Solution                                                                              |
|------|--------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports a single/local user.                                               | Deploy on cloud platforms such as Render, AWS, or Azure to support multiple users simultaneously.        |
| 2    | Data Storage       | Uses a CSV file for training data and Pickle files (model.pkl, scaler.pkl). | Integrate a database such as MySQL or MongoDB to store applicant details and prediction history.         |
| 3    | Performance        | Prediction is performed using a locally loaded model.                       | Optimize the model, enable caching, and use a production server (Gunicorn or uWSGI) for faster response. |
| 4    | Security           | No authentication and limited input validation.                             | Add user authentication, HTTPS, input validation, and role-based access control.                         |

Future Roadmap:

| Phase   | Planned Feature / Enhancement                                           | Target Timeline | Expected Impact                                              |
|---------|-------------------------------------------------------------------------|-----------------|--------------------------------------------------------------|
| Phase 1 | Deploy the Flask application to a cloud platform (Render/AWS/Azure).    | 1 Month         | Allow online access from any location.                       |
| Phase 2 | Integrate a database to store applicant details and prediction history. | 2 Months        | Improve data management and record keeping.                  |
| Phase 3 | Implement user login, authentication, and role-based access control.    | 3 Months        | Enhance application security and access management.          |
| Phase 4 | Improve the machine learning model using larger datasets and features . | 4–6 Months      | Increase prediction accuracy and overall system reliability. |